Crane Wire Rope Inspection Checklist (29 CFR 1926.1413)

For shift, monthly and annual wire rope inspections on cranes and derricks under Subpart CC.

OSHA 29 CFR 1926.1413

DATE	SHIFT
INSPECTOR	AREA / EQUIPMENT ID

Before you start

1. Mark each item OK, Defect or N/A.
2. Report any Defect at once to the competent or qualified person running the inspection; 1926.1413(a)(4) and (c)(3) set when the rope must come out of service.
3. The rope manufacturer's instructions and site rules may add checks.

Legal basis: 29 CFR 1926.1413: Wire rope—inspection (Part 1926 Subpart CC, Cranes and Derricks in Construction) Source: eCFR, Title 29, text as of 2026-11-09

Online version and sources: https://safetyindex.blog/us/standards/1926-1413-wire-rope-inspection

SECTION A | Shift inspection and critical review items

No.	Check item	What to look for	OK	Defect	N/A	Notes
1	Shift inspection timing Ref. 1926.1413(a)(1)	Competent person began the visual inspection before the shift and finished it during or before the shift	☐	☐	☐	
2	Ropes covered Ref. 1926.1413(a)(1)	All running and standing ropes likely to be used this shift were observed	☐	☐	☐	
3	Rotation resistant rope Ref. 1926.1413(a)(3)(i)	Rotation resistant rope in use was examined closely	☐	☐	☐	
4	Boom hoist and luffing hoist rope Ref. 1926.1413(a)(3)(ii)	Rope checked along its length, especially at reverse bends	☐	☐	☐	
5	Drum points Ref. 1926.1413(a)(3)(iii)	Rope at flange, crossover and repetitive pickup points on drums checked	☐	☐	☐	
6	Terminal ends Ref. 1926.1413(a)(3)(iv)	Rope at or near terminal ends checked	☐	☐	☐	
7	Saddles and sheaves Ref. 1926.1413(a)(3)(v)	Rope touching saddles, equalizer sheaves or limited-travel sheaves checked	☐	☐	☐	

SECTION B | Category I and Category III deficiencies

No.	Check item	What to look for	OK	Defect	N/A	Notes
8	Rope structure Ref. 1926.1413(a)(2)(i)(A)	No significant kinking, crushing, unstranding, birdcaging, core failure signs or steel core protruding between outer strands	☐	☐	☐	
9	Corrosion Ref. 1926.1413(a)(2)(i)(B)	No significant corrosion on the rope	☐	☐	☐	
10	Arc and heat damage Ref. 1926.1413(a)(2)(i)(C)	No electric arc damage from a non-power-line source and no heat damage	☐	☐	☐	
11	End connections fitted Ref. 1926.1413(a)(2)(i)(D)	End connections correctly applied	☐	☐	☐	

General guidance only, not legal advice. Check the current regulation and your employer's procedures.

SECTION B | Category I and Category III deficiencies

No.	Check item	What to look for	OK	Defect	N/A	Notes
12	End connection condition Ref. 1926.1413(a)(2)(i)(E)	End connections not significantly corroded, cracked, bent or worn	☐	☐	☐	
13	Rotation resistant core Ref. 1926.1413(a)(2)(iii)(A)	No core protrusion or other distortion showing core failure	☐	☐	☐	
14	Power line contact Ref. 1926.1413(a)(2)(iii)(B)	Rope has had no prior electrical contact with a power line	☐	☐	☐	
15	Strands Ref. 1926.1413(a)(2)(iii)(C)	No broken strand anywhere on the rope	☐	☐	☐	

SECTION C | Category II broken wires and diameter

No.	Check item	What to look for	OK	Defect	N/A	Notes
16	Running ropes Ref. 1926.1413(a)(2)(ii)(A)(1)	Fewer than six random broken wires per lay and fewer than three broken wires in any one strand per lay	☐	☐	☐	
17	Rotation resistant ropes Ref. 1926.1413(a)(2)(ii)(A)(2)	Fewer than two random broken wires per six rope diameters and fewer than four per 30 rope diameters	☐	☐	☐	
18	Pendants and standing ropes Ref. 1926.1413(a)(2)(ii)(A)(3)	No more than two broken wires per lay beyond end connections and no more than one per lay at an end connection	☐	☐	☐	
19	Rope diameter Ref. 1926.1413(a)(2)(ii)(B)	Reduction from nominal diameter is 5% or less	☐	☐	☐	

SECTION D | Removal from service and repair

No.	Check item	What to look for	OK	Defect	N/A	Notes
20	Category I hazard decision Ref. 1926.1413(a)(4)(i)	Competent person made an immediate safety-hazard determination for each Category I finding	☐	☐	☐	
21	Category II rope out of use Ref. 1926.1413(a)(4)(ii)	Rope not used until the manufacturer's removal criterion is met, the rope is replaced, or a localized section is severed out	☐	☐	☐	
22	Category III rope out of use Ref. 1926.1413(a)(4)(iii)	Rope not used until replaced, or until a localized defect other than power line contact is severed out	☐	☐	☐	
23	No splices Ref. 1926.1413(a)(4)(i)(B)	No lengths of rope joined by splicing	☐	☐	☐	
24	Power line contact repair Ref. 1926.1413(a)(4)(iii)(B)	Rope that touched an energized power line has not been repaired	☐	☐	☐	
25	Drum wraps after shortening Ref. 1926.1413(a)(4)(i)(B)	Two wraps remain on the drum with the load and/or boom at its lowest position	☐	☐	☐	
26	Tag-out Ref. 1926.1413(a)(4)(iv)	Equipment or the affected hoist tagged out per 1926.1417(f)(1) until the rope is repaired or replaced	☐	☐	☐	

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SECTION E | Monthly, annual, lubricant and records

No.	Check item	What to look for	OK	Defect	N/A	Notes
27	Monthly inspection Ref. 1926.1413(b)(1)	Inspection to paragraph (a) done this month	☐	☐	☐	
28	Monitored deficiencies Ref. 1926.1413(b)(2)	Deficiencies flagged for monitoring at the annual inspection were checked	☐	☐	☐	
29	Monthly record Ref. 1926.1413(b)(4)	Monthly inspection documented per 1926.1412(e)(3)	☐	☐	☐	
30	Annual inspection Ref. 1926.1413(c)(1)	Qualified person inspected ropes in use to paragraph (a) within the last 12 months	☐	☐	☐	
31	Full-length inspection Ref. 1926.1413(c)(2)(ii)	Complete and thorough check of the rope surface along its entire length done within the 12-month period	☐	☐	☐	
32	Annual record Ref. 1926.1413(c)(4)	Annual/comprehensive inspection documented per 1926.1412(f)(7)	☐	☐	☐	
33	Lubricant Ref. 1926.1413(d)	Lubricant on the rope does not hinder inspection	☐	☐	☐	
34	Document access Ref. 1926.1413(e)	Inspection documents available to all inspectors during the retention period	☐	☐	☐	

DEFECTS FOUND AND ACTION TAKEN

Item no.	Defect description	Reported to	Action taken / date closed

SIGN-OFF

SAFE TO OPERATE? ☐ Yes ☐ No: tagged out of service	
INSPECTOR SIGNATURE	DATE / TIME
SUPERVISOR REVIEW	DATE / TIME